

Creating Cohorts of Songs

Spotify Audio Features - Rolling Stones Dataset

Machine Learning Course-End Project Report

Domain	Music Streaming / ML
Dataset	Rolling Stones - Spotify API
Technique	K-Means Clustering + PCA
Dataset Size	1,610 songs (before Cleaning)

Note: This report was generated with assistance of AI.

1. Executive Summary

Spotify, with over 456 million active monthly users, relies on personalized recommendations to keep users engaged. This project applies unsupervised machine learning to group Rolling Stones songs into distinct cohorts based on Spotify audio features. The resulting clusters can power a recommendation engine that suggests contextually similar songs to listeners.

Using K-Means clustering ($k=4$) on 11 standardized audio features, after outlier removal via Z-score - four clearly interpretable song cohorts were identified: Popular High-Energy Studio Hits, Mellow Acoustic Chill Studio Tracks, Live Performances/Live High Energy Rock Tracks, and Groovy Upbeat Studio Tracks.

2. Problem Statement & Objective

The customer always looks forward to specialized treatment, whether shopping over an e-commerce website or watching Netflix. Spotify intends to create cohorts of different songs that will aid in the recommendation of songs to users based on various relevant features. Each cohort would contain similar types of songs.

Objective

As a data scientist, perform exploratory data analysis and cluster analysis to create cohorts of songs. The goal is to gain a better understanding of the various factors that contribute to creating a cohort of songs.

3. Dataset Description

The dataset contains Spotify API data for all Rolling Stones albums listed on Spotify. Each row represents one unique song identified by a Spotify track ID.

Feature		Description
acousticness	0 – 1	Confidence measure of whether track is acoustic
danceability	0 – 1	Suitability for dancing based on tempo, rhythm

energy	0 – 1	Perceptual measure of intensity and activity
instrumentalness	0 – 1	Predicts whether track contains no vocals
liveness	0 – 1	Detects presence of live audience in recording
loudness	-60 to 0 dB	Overall loudness of track in decibels
speechiness	0 – 1	Presence of spoken words in the track
tempo	BPM	Estimated tempo in beats per minute
valence	0 – 1	Musical positiveness - high = happy/cheerful
popularity	0 – 100	Spotify popularity score based on play counts
duration_ms	milliseconds	Track duration in milliseconds

4. Data Preprocessing & Cleaning

4.1 Initial Observation

The raw dataset contained 1,610 rows and 18 columns. Initial inspection revealed:

- No missing values across all columns
- An unnamed index column (Unnamed: 0) was dropped
- No Duplicate records are found, if any duplicate rows and duplicate Spotify track IDs were removed, retaining the first occurrence
- release_date was parsed to datetime and new feature release_year created

4.2 Feature Scaling - StandardScaler

StandardScaler was applied to all 11 numeric audio features before clustering.

- loudness ranges from -60 to 0 dB
- tempo ranges from 0 to 250+ BPM
- Most audio features (acousticness, energy etc.) range from 0 to 1

Without scaling, loudness and tempo would dominate due to high value and affects K-Means, producing biased clusters.

4.3 Outlier Removal - Z-Score

After scaling, Z-score was applied to remove statistical outliers. Any song where any feature value exceeded 3 standard deviations from the mean was removed.

Threshold: $|Z\text{-score}| < 3$ for all features simultaneously. The dataset after cleaning contained approximately 1,508 songs.

5. Exploratory Data Analysis

5.1 Album Recommendations

To identify the top 2 albums to recommend to any user, albums were ranked by the mean of count of songs with popularity score ≥ 40 :

Rank	Album	Popular Songs (score ≥ 40)
1	Sticky Fingers (Remastered)	47.857143
2	Exile On Main Street (2010 Re-Mastered)	43.058824
3	Some Girls	42.500000
4	Tattoo You (2009 Re-Mastered)	41.777778
5	Beggars Banquet (50th Anniversary Edition)	40.444444

5.2 EDA Findings

- Energy and loudness are strongly positively correlated ($r = 0.76$) - louder songs tend to be more energetic
- Acousticness and energy are strongly negatively correlated - acoustic songs tend to have lower energy
- Popularity shows no strong linear correlation with any single feature - it depends on a combination
- Songs released in the 1970s–1980s tend to have higher popularity today, indicating classic catalog dominance
- Live recordings form a natural distinct group - high liveness + high energy + long duration

5.3 Dimensionality Reduction - PCA

Principal Component Analysis (PCA) was applied to understand the structure of the 11-dimensional feature space.

- 11 principal components were computed
- Approximately 8 components explain 95% of total variance

PCA serves two purposes in this project: (1) Understanding which features contribute most to variation, and (2) Projecting songs into 2D space for cluster visualization.

6. Cluster Analysis

6.1 Identifying the Right Number of Clusters - Elbow Method

K-Means was run for $k = 2$ through 12. Inertia (Within-Cluster Sum of Squares) was recorded for each value of k and plotted as the Elbow Curve.

k	Inertia	Inertia Drop
2	11736	N.A
3	10449	1287
4	9570	911
5	9036	534
6	8388	648
7	7953	435

The elbow at $k=4$ indicates that adding more clusters beyond 4 produces diminishing improvements in compactness. $k=4$ was chosen as the optimal number of clusters.

6.2 Algorithm - K-Means

K-Means clustering was selected for the following reasons:

- Efficient and scalable for datasets of this size (1,508 songs)
- Produces compact, well-separated clusters suitable for a recommendation engine

6.3 Cluster Definitions

The mean feature values per cluster were computed on the original (pre-scaled) data for interpretability. The heatmap is used to assign meaningful labels to each cluster.

Cluster	Label	Size	Dominant Features
0	Popular High-Energy Studio Hits	356 songs	Highest popularity (~27.3), loudest (-4.9 dB), high instrumentalness (0.24), high valence(0.74)
1	Acoustic / Mellow Ballads	262 songs	Highest acousticness (0.48), lowest energy (0.59), sad/emotional tone (valence 0.39)
2	Live Performances	520 songs	Highest liveness (0.84), fastest tempo (137 BPM), longest duration (280s), high speechiness (0.1)
3	Groovy Upbeat Studio Tracks	361 songs	Highest danceability (0.60), highest valence (0.75), shortest duration (187s)

CLUSTER 0 - POPULAR HIGH-ENERGY STUDIO HITS

This cluster contains the most commercially successful Rolling Stones studio recordings. Songs here have the highest average popularity score (~27.3) and are the loudest and most energetically charged tracks in the catalog. These are typically the well-known radio hits that have accumulated the most streams on Spotify.

Representative Songs: Honk (Deluxe)

Recommended Use: Greatest Hits playlists, Rock workout playlists

CLUSTER 1 - MELLOW ACOUSTIC CHILL STUDIO TRACKS

Characterized by the highest acousticness (0.48) and the lowest energy of all four clusters. These tracks have a melancholic, emotional tone reflected in the lowest valence score (0.39). The slower tempo (107 BPM average) and longer duration indicate rich, fully developed compositions.

Representative Songs: Their Satanic Majesties Request (50th Annivers..)

Recommended Use: Late-night playlists, Mood/chill playlists, Ballad lovers

CLUSTER 2 - LIVE HIGH ENERGY ROCK TRACKS

This cluster is dominated by live concert recordings, clearly indicated by the highest liveness score (0.84) by a large margin. The raw energy of live performance is captured in the highest overall energy (0.93) and fastest tempo (137 BPM). Extended live versions explain the longest average duration (280 seconds). Lower popularity (17) reflects that listeners tend to stream studio versions over live recordings.

Representative Songs: Voodoo Lounge Uncut (Live)

Recommended Use: Concert experience playlists, Live music enthusiasts

CLUSTER 3 - GROOVY UPBEAT STUDIO TRACKS

These are the most danceable (0.60) and most positive-sounding songs (valence 0.75) in the catalog. Short and punchy with an average duration of just 187 seconds, these tracks represent the tighter, groove-focused side of the Rolling Stones' studio work. Lower popularity (15) suggests these are deeper cuts rather than mainstream hits.

Representative Songs: Aftermath (UK Version)

Recommended Use: Dance/party playlists, Deep cuts for dedicated fans

7. Conclusions & Recommendations

Conclusions

- Four distinct song cohorts were successfully identified using K-Means clustering
- Top 2 recommended albums: Exile On Main Street (14 popular songs) and Sticky Fingers (10 popular songs)
- Live recordings form a naturally distinct group and should be treated separately in recommendation pipelines
- Popularity is highest in the High-Energy Studio Hits cluster - these are the best songs for onboarding new listeners

Recommendations

- Use cohort labels as a feature in collaborative filtering models to improve cold-start recommendations
- Serve Cluster 0 (High-Energy Hits) to new users as a default entry point
- Build mood-based playlists: Cluster 1 for late night, Cluster 3 for parties, Cluster 2 for concert fans
- Retrain clusters periodically as new albums and streaming data become available
- Extend analysis to full Spotify catalog for cross-artist recommendations
-